BE - Mini projects

Objective of the BE

The objective of this BE is to give the opportunity to students to apply the theory studied during the Lectures, to practical issues in Reliability theory or in Actuarial sciences for instance.

More precisely, regarding the problem considered, the students are required to:

1. Present the studied mathematical framework, which matches the particular situation under interest.
2. Provide some mathematical material (and analysis) related to the objective of the problem.
3. Perform a numerical analysis of the results.
4. Give an interpretation of the theoretical and numerical results.
5. Suggest additional directions for possible future analysis.

Agenda

The BE work is composed of:

1. 3 working time slots in the Agenda (see ADE).
2. Notebook/Rmarkdown deadline on **Tuesday May 21st, at midnight**.
3. Defense sessions on May 24th.

Organisation

Students will work in **8 groups** composed of 3 students. Please, complete the form available at

`https://lite.framacalc.org/be-poisson-4ma-a6og`

with your first names and surnames (1 line per group). Each group will be assigned one project out of the presented ones. Since the supervising manpower is limited, some project will be handled by two different groups.

- **Assessment:**

Each group is due to:

1. Deliver on the Moodle page a notebook iPython or a R Markdown containing the results (written with LaTeX) and the numerical results.
2. Perform a defense on the project.

- **Expected work:**

1. The Notebook iPython or the R Markdown (as specified for each project) will contain:
   (a) a presentation of the main issue of the project;
   (b) a description of the mathematical model associated to it;
   (c) the theoretical treatment of the main issues;
   (d) a clear presentation of the simulations performed and the numerical results;
   (e) an interpretation of the results;
   (f) a conclusion, including future lines of possible investigation.
2. The defense will be performed in French or in English by the three members of the group (who will contribute equally). It is aimed to last 20-25 min per group (including a 15 min presentation and a 5-10 min discussion time). During the defense, students must present the main issue of the project, the main results (theoretical and numerical) together with their interpretation(s). They are encouraged to use slides during their oral presentation.

The supervisors encourage initiatives from the students to study additional or different questions in lines with the main topic of the project.